Code-backed research, 2025–2026

What the recent literature changes in the platform

Window	2025-01-01 to 2026-10-13
Harvested	1 639 unique arXiv papers
With resolvable code	331
After filtering	**45**
Reproduce	`make research`

Every repository was resolved through the GitHub API on the generation date: it exists, and its licence, size, stars and last push are recorded as returned.

No repository was executed. “Has code” means the code exists, not that it reproduces the paper.

Branch `research/code-backed-deep-dive`. Tooling: `research/tools/harvest_arxiv_code.py`, `research/tools/curate.py`. Raw harvest committed under `research/data/`.

The question, and why it is different from “relevant papers”

The upstream `resources/academic_papers.md` lists 32 papers chosen for *relevance*. Right first filter, wrong question for a build team.

This asks something narrower and verifiable:

Of the recent condition-monitoring literature, which papers come with code we can read, run and — licence permitting — reuse?

That is a question about *artefacts*, not ideas. It has an answer you can check.

6.3 %

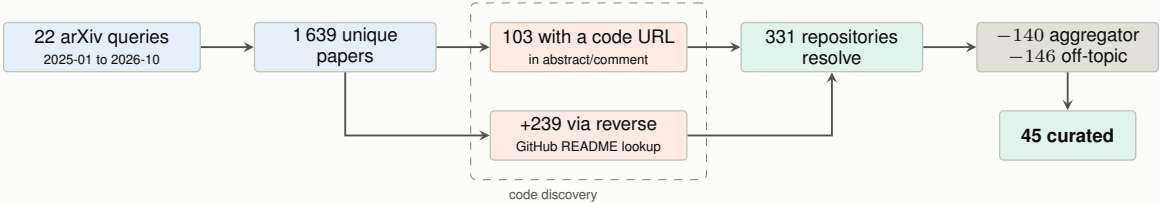
of papers print a code URL in the abstract or comment

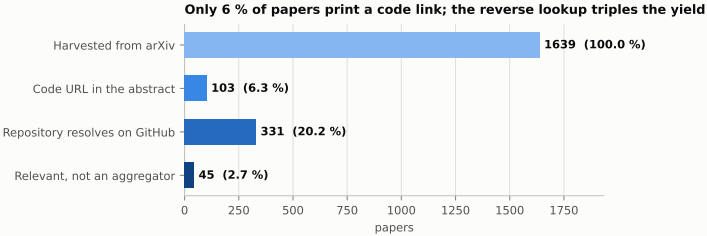
27 / 45

of the code-backed papers ship code with *no usable licence*

So what: Relevance filters find ideas. Artefact filters find things you can build on this afternoon — and most of them, legally, you cannot.

Pipeline





Reverse lookup: `GET/search/repositories?q="<arxiv_id>"+in:readme` — 900 searches spent, 239 matched (27 %), capped at 30 requests/minute even authenticated.

The filter that the first version of this needed

The reverse lookup over-reports in one systematic way, and the first run was visibly wrong because of it.

A whole class of repositories — `*-arxiv-daily`, `awesome-*`, paper lists — cites *thousands* of arXiv IDs in its README.

The first run therefore ranked:

- `BaiShuanghao/my_arXiv_daily` (217⋆) as the code for **eight** unrelated papers
- `halsay/ASR-TTS-paper-daily` (513⋆) as the code for a paper on analog resonant networks

Two filters remove them:

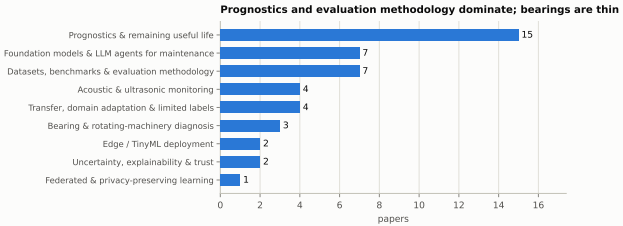
1. Name and description patterns — `arxiv-daily`, `paper-daily`, `awesome-`, `^arxiv-`, `*-list`, `literature`.

2. The robust one: any repository cited by ≥ 3 different papers through README search. Real code is cited by one paper, sometimes two when a lab reuses a repository. Three is already an aggregator.

54 repositories dropped $\Rightarrow$ 140 papers whose only “code” was a listing.

So what: An automated literature harvest needs an adversarial filter, and the data-driven signal (how many papers does this repo claim?) beats the pattern match.

Where the code-backed work actually is



Two things stand out, and both are uncomfortable.

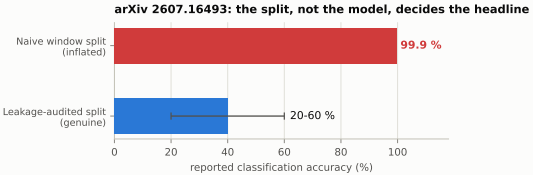
Prognostics and evaluation methodology dominate. The field is arguing about *how it measures*, not which architecture wins.

Bearings are thin — 3 of 45. The serious rotating-machinery literature is in *MSSP*, *IEEE TIM* and the PHM Society conferences, which are not on arXiv and do not ship code.

Acoustic emission: zero. Not one code-backed AE paper in 21 months.

So what: This list is biased toward the ML-facing half of the field. Know that before using it to plan hardware.

Shift 1 — evaluation methodology became the finding



2607.16493 (CALIBRA, MIT) on C-MAPSS, NASA IMS and UCI Hydraulic: naive sliding-window splitting inflates accuracy **from a genuine 20–60 % to 99.9 %** — or collapses it to 0 % through degenerate class representation.

It is a cluster, not one corrective paper:

2605.28345 **Picid** (EPFL-IMOS) — evaluation *infrastructure* that enforces leakage-safe, deterministic dataset construction. A lab thought this was worth building.

2607.08273 — bearing RUL with held-out units defined by **load–speed regime**, reporting nMAE 0.1477 *with* empirical 90 % interval coverage.

2606.29339 — separates **sensor-failure tolerance** from **low-SNR robustness**, two things routinely conflated, over Hi-net seismic + FORGE borehole DAS + MAFAULDA vibration.

So what: The gap between a leaked and an honest split is larger than the gap between any two architectures in this window. Protocol discipline buys more than model selection.

Shift 2 — foundation models became a frozen feature extractor



2606.11990 (EUSIPCO 2026) — **Chronos-2 frozen** + a small regression head consistently beats recurrent baselines on real industrial sensor data, with no feature engineering and no large labelled set.

2601.22631 **PEFT-MuTS** — few-shot RUL via parameter-efficient fine-tuning, and explicitly "contrary to the widely held view": transfer does *not* need similar devices.

2604.20122 — adaptive conformal anomaly detection on `ibm-granite/granite-tsfm` (910⋆, Apache-2.0) — maintained, permissive infrastructure rather than a research dump.

What we are assuming, stated plainly

The backbone consumes our *feature frames* at one frame per acquisition — **not** 25.6 kHz waveforms. No paper in this window demonstrates a TSFM on raw high-rate vibration.

It does **not** replace the unsupervised path: a frozen backbone still needs labels for its head, and a new site has none. This is a Phase-2 capability, behind the unsupervised-first commitment.

Shifts 3–5

3. Calibration is standard now

- 2607.08273 forms RUL intervals by empirical residual calibration and reports *coverage*, not just MAE.
- 2604.20122 applies adaptive conformal prediction — distribution-free coverage that survives drift.
- 2511.12240 **SCI** wraps a model in a closed entropy-gated loop: stop, continue, or escalate.
- 2606.12486 (Scania) builds on wear being **monotonically non-decreasing** — a physics prior that doubles as a calibration check.

4. INT4 is defensible

2604.08474: a 9 697-parameter 1-D CNN, federated on C-MAPSS, quantised at 32/8/4/2 bits over **10 seeds**.

INT4 is **statistically indistinguishable** from FP32 ($p = 0.341$ FD001, $p = 0.264$ MAE FD002) at $8\times$ reduction.

Multi-seed with p-values is more rigorous than this literature's norm, and the model size matches ours (Deep SVDD $\approx$1.7 k params).

2606.24173 keeps *classical* baselines in the table alongside accuracy, size and CPU latency. Right shape for our per-kit tables.

5. Adaptation went source-free

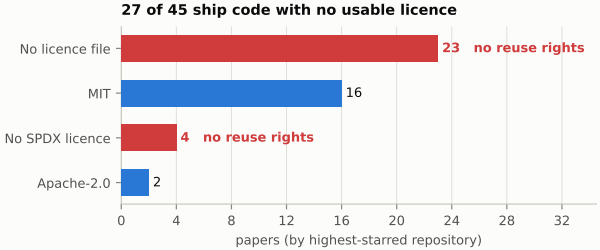
Both shifts driven by one constraint: *you cannot move the data*.

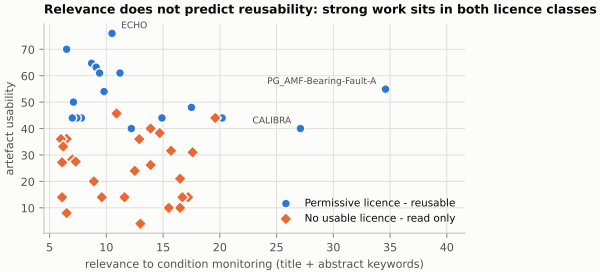
2503.08749 **SDALR** (41$\star$) — adapt a bearing model to a new domain with **no access to source data**. Lets customer A's model serve customer B without either's data leaving site.

2609.35792 — **matched-protocol** comparison of ring gossip vs FedAvg vs local vs centralised: one implementation, architecture, split and budget. Removes the aggregation server entirely.

So what: Source-free adaptation across tenants; federation or gossip within a tenant. "Customer data never joins another customer's training round" is a far easier claim to defend for FedRAMP.

The constraint nobody mentions in a literature review





A public GitHub repository with no `LICENSE` file is **all rights reserved**. You may read it and learn the method. You may not copy a function out of it.

And the most *useful* repositories here are frequently the unusable ones:

- `BdLab405/SDALR` (41★, the best-regarded on-topic bearing repo in the window) — **no licence**
- `EPFL-IMOS/picid` (the evaluation infrastructure we most want) — **NOASSERTION**

Mitigation that actually works: open an issue asking for a licence. Most academic authors intend reuse and simply did not add the file. Costs an issue and a week.

Adopt now — permissive, relevant, substantial

Repository	Licence	What we take, and why
nbbllxx0/Empirical-Calibration-\protect\global\let\T1\textellipsis.\kern\fontdimen3\font.\kern\fontdimen3\font.\kern\fontdimen3\font\T1\textellipsis	MIT	**Highest-priority read of the list.** Closest paper to our actual problem: bearings, time-varying load and speed, 10 documented units, calibrated intervals. Take the regime-based split and the empirical residual calibration.
therealdeadbeef/aerospace-fl-quantization	MIT	The multi-seed quantisation protocol, not the model. Makes every per-kit INT8/INT4 claim defensible.
ISA09/cephalon-v2	MIT	Two-axis robustness test design: channel dropout and SNR degradation as *separate* reported conditions. We have no sensor-failure test at all today.
rajeevfee/PG_AMF-Bearing-Fault-Analysis-	MIT	Learned adaptive moment features vs our 14 fixed time-domain indicators. Parameters are flash constants either way — no runtime penalty if it wins.
TakuyaFujimura/dcase-asd-toolkit	MIT	Reference implementation for the acoustic channel; a toolkit, not a one-paper dump.
ZhixiangWang-CN/gossip-lstm-cmapss	MIT	The matched-protocol harness. The only honest way to decide federated-vs-alternatives.
ibm-granite/granite-tsfm	Apache-2.0	Adaptive conformal thresholds — replaces our parametric Gumbel tail extrapolation.

Full triage with sequencing and costs: `research/deep_dive/03_adoption_shortlist.md`. Licences as returned by the GitHub API on 2026-10-13 — re-check before copying anything.

What this changes in the seven product plans

Plan	Severity	Change
00 Overview	**High**	Evaluation protocol becomes a phase-*exit gate*: named split, zero leakage violations, regime-level held-out units, the two robustness axes reported separately, metrics quoted at an agreed alarm budget.
01 Sensor HW	Medium	Keep AE as the hardware differentiator and keep the front-end on the PCB, but move AE *algorithms* out of v1: zero code-backed AE papers and no public AE benchmark to validate against. Ship the 6 kHz MEMS path first.
02 Edge firmware	Medium	Add an INT4 evaluation path. Replace schedule-driven tiering with *uncertainty-gated* escalation (cheap indicators → spectrum → envelope → raw snapshot).
03 IoT pipeline	—	No architectural change. Add per-device sequence-gap accounting: a sensor dropping 30 % of frames looks healthy on a trend chart.
04 ML/AI engine	**High**	Frozen-TSFM + small head as a Phase-2 track. Conformal intervals replace parametric tail extrapolation. Enforce monotonicity on the health index.
05 Cloud platform	Medium	Source-free adaptation across tenants; federation or gossip within a tenant.
06 CMMS dashboard	Medium	Abstention and calibrated confidence become first-class UI states. RUL renders as a distribution, never a point.
07 Electrical	Low	One code-backed MCSA paper in 21 months. Keep the plan, lower the stated confidence, budget for an *own* measurement campaign rather than literature transfer.

Full reasoning per plan: `research/deep_dive/02_gap_analysis_vs_plans.md`.

Three things not to do, and what did not change

Do not

1. Benchmark on C-MAPSS. Nine of 45 entries use it. Simulated turbofan, no vibration channel, saturated. It would make our results comparable to the literature and uninformative about the product.

2. Vendor unlicensed code. 27 of 45. Read them; do not copy them.

3. Chase architecture. The leaked-vs-honest split gap (99.9 % vs 20–60 %) exceeds any architecture gap in this window.

What did not change

- **Envelope analysis and spectral kurtosis remain the technique of record** for bearing faults. Nothing displaces them; 2606.26317 generalises adaptive moments but works inside exactly that framework.
- **Unsupervised one-class detection is still the only day-one option.** No paper here gets around the absence of labelled failures at a new site.
- **Nobody solved cross-machine generalisation.** Source-free adaptation, PEFT and TSFM embeddings all *reduce* the target data needed. None removes it.
- **The gap that matters is data, not method:** no public dataset covers a wireless MEMS sensor on a real industrial machine over a multi-month degradation. That is our pilot, and it is on the critical path.

Reproducing this

```
# ~45 min: 22 arXiv queries, 103 repo lookups, 900 README searches.
# Standard library only -- runs in CI and on a gateway, no virtualenv.
GITHUB_TOKEN=<public-read> python3 research/tools/harvest_arxiv_code.py \
    --out research/data/harvest_$(date +%Y%m%d).json \
    --from 2025-01-01 --max-per-topic 150 --readme-search-budget 900

python3 research/tools/curate.py --in research/data/harvest_$(date +%Y%m%d).json \
    --out-md research/code_backed_papers_2025_2026.md \
    --out-csv research/data/papers_with_code.csv \
    --out-json research/data/curated.json

python3 slides/figures/make_figures.py   # figures come from the committed data
```

Limitations, stated up front. arXiv only — *MSSP*, *IEEE TIM* and the PHM Society conferences carry much of the serious condition-monitoring literature and are absent. English abstracts only. README-search recall is bounded by GitHub's indexing. A single snapshot: star counts and push dates aged the moment it ran. The relevance threshold (6.0) was calibrated by hand against this harvest, and both failure directions are recorded with examples in the audit trail of the generated list, so it can be re-argued rather than taken on trust.